Securing CCTV Cameras Against Blind Spots

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About Me

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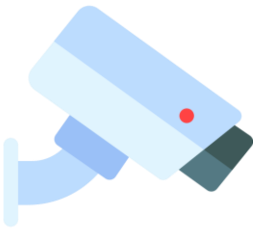
Motivation

In recent years, neural networks have been deployed in various real-world AI systems.

These systems demonstrate high performance in their tasks (e.g., person/product identification).

However, many attacks targeting the security of these systems are developed and used to accomplish various goals.



Motivation

In many cases, the model output has financial and security implications (e.g., sending bills, charging for products, identifying people in airport terminals/on the street, etc.).

These implications can motivate making objects and/or people disappear, avoiding detection by these systems.

- Financial – save money/avoid payment.
- Security – avoid detection/tracking.

Model Evasion Attacks

We have seen model evasion attacks, targeting a model's object/person detection capabilities, performed against:

- Facial Recognition models [1].
- Autonomous vehicles [10].
- Object detectors [15].

(a) (b) (c) (d)

Figure 4: Examples of successful impersonation and dodging attacks. Fig. (a) shows S_A (top) and S_B (bottom) dodging against DNN_B. Fig. (b)–(d) show impersonations. Impersonators carrying out the attack are shown in the top row and corresponding impersonation targets in the bottom row. Fig. (b) shows S_A impersonating Milla Jovovich (by Georges Biard; source: https://goo.gl/GlsWlC); (c) S_B impersonating S_C; and (d) S_C impersonating Carson Daly (by Anthony Quintano; source: https://goo.gl/VfnDct).

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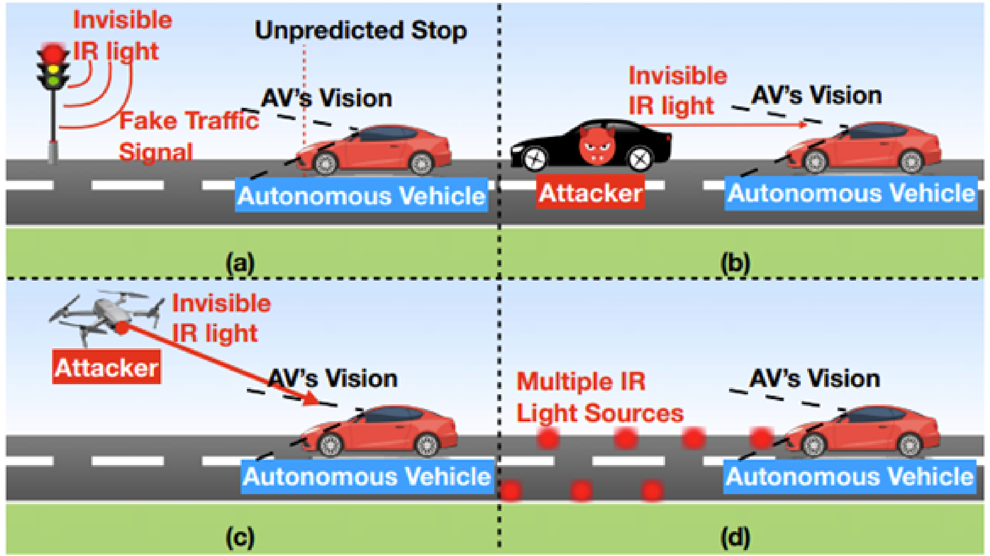

Figure 5: (a) The attacker creates fake invisible traffic signal; (b) The attacker uses IR light to alter AV's environment perception results; (c) The attacker can blind the AV's camera to create frequent system alters; (d) The attacker controls the IR sources on the road to alter AV's SLAM results.

Model Evasion Attacks

We have seen model evasion attacks, targeting a model's object/person detection capabilities, performed against:

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- Autonomous vehicles [10].
- Object detectors [15].

Fig. 1: In this demonstration, the YOLOv2 detector is evaded using a pattern trained on the COCO dataset with a carefully constructed objective.

Model Evasion Attacks

We found that evasion attacks on object detectors leverage:

- Infrared (IR) light [2,10].
- LED light saturation [8].
- Angles [3,4,6,7,11].
- Adversarial patches [1,3-5,7,9,11-15].

Model Evasion Attacks

We found that evasion attacks on object detectors leverage:

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- LED light saturation [8].
- Angles [3,4,6,7,11].
- Adversarial patches [1,3-5,7,9,11-15].

Can we leverage these factors in an evasion attack which is only dependent on the location of the evading person, relative to the object detector?

Analysis – Location Effect on Object Detectors

We perform two analyses on object detectors' location-based behavior:

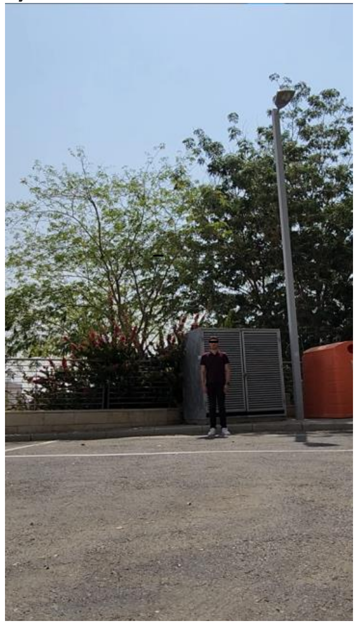
1. Object detector confidence when varying a single person's distance, angle, and height from the observing camera.
2. Object detector confidence when observing a high pedestrian traffic area over time.

Analysis – Distance/Angle Effect on Object Detectors

We investigate the effect of a person’s distance and angle, and the quality of the footage fed to an object detector, on the object detector’s confidence.

Experimental Setup: We analyzed a person standing at three distances (2.5m, 5m, 10m), and eight angles (0°, 45°, 90°, 135°, 180°, 225°, 270°, 315°), from a 1080p camera positioned at three heights (0.6m, 1.8m, 2.4m).

Height: 0.6m
Angle: 0°



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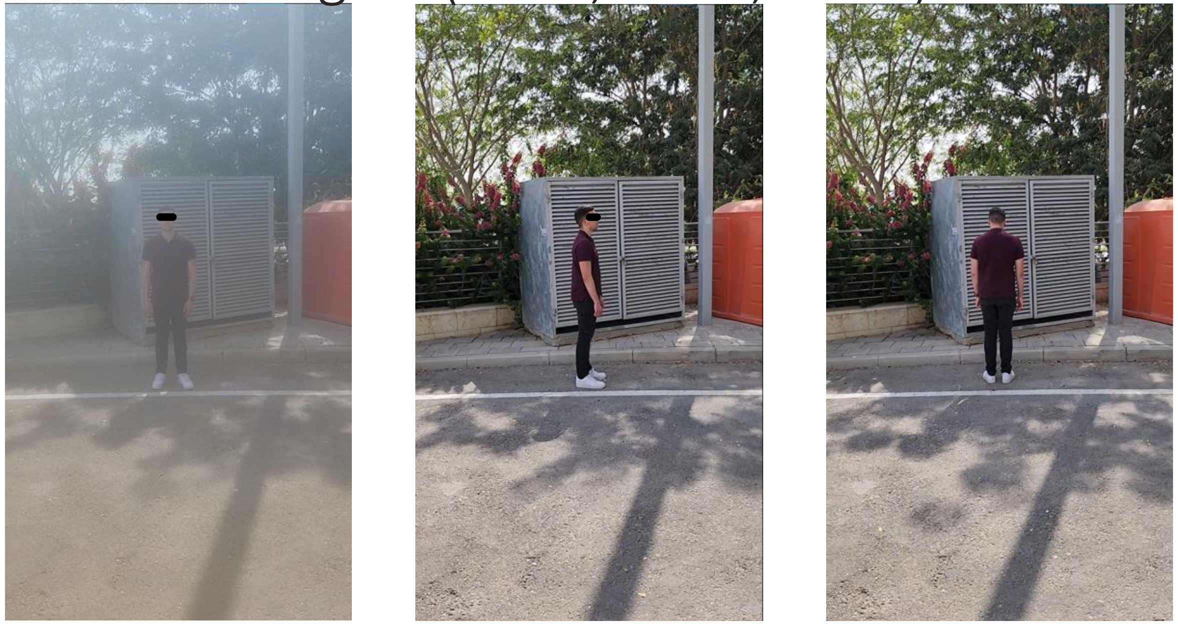
Height: 1.8m
Distance: 5m



Analysis – Distance/Angle Effect on Object Detectors

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Distance: 10m
Angle: 0°

Analysis – Distance/Angle Effect on Object Detectors

For each position (height, angle, distance) we recorded the person standing still for one minute, and fed the footage to an object detector.

For each video:

- We recorded the confidence values of all the bounding boxes labelled “person“ by the object detector.
- We then calculated the mean confidence of the model’s “person” detection bounding boxes.

We report, for each position, the mean confidence values of bounding boxes labelled “person”.

In other words, given a person’s position relative to an object detector, how confident is the object detector when detecting the person?

Analysis – Distance/Angle Effect on Object Detectors

We repeated this analysis with five different object detectors:

YOLOv3, Faster R-CNN, SSD, DiffusionDet, RTMDet.

We repeated this analysis in four different lighting conditions:

- Laboratory environment.
- Morning.
- Afternoon.
- Night.

Analysis – Distance/Angle Effect on Object Detectors (YOLOv3)

Results (Camera Resolution: 1080x1920):

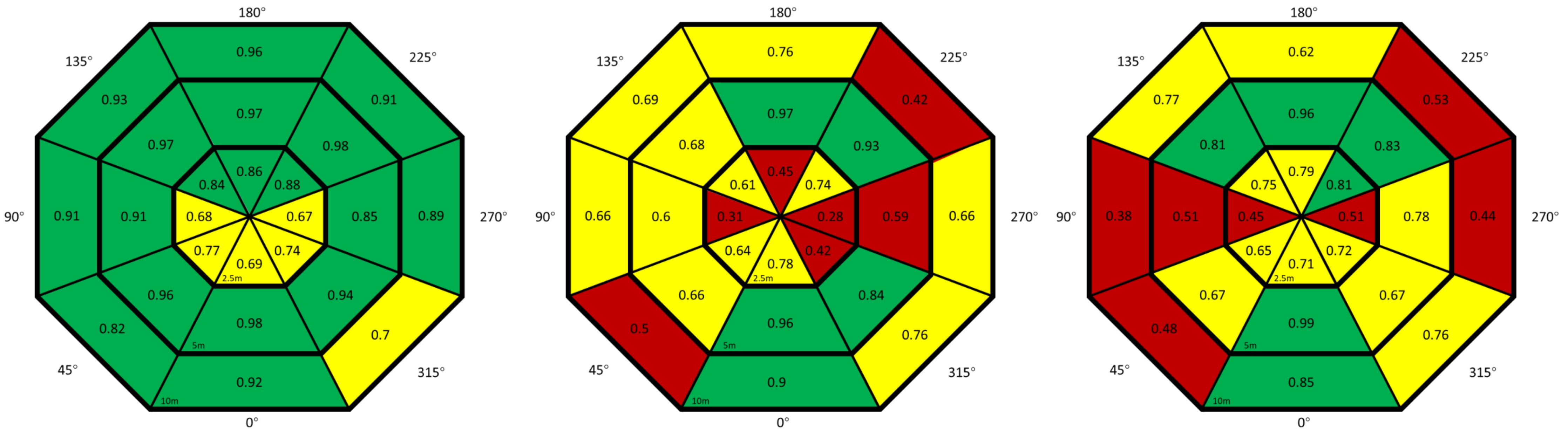
Legend

> 0.8

0.6 – 0.8

< 0.6

Height: 0.6m 1.8m 2.4m



Analysis – Distance/Angle Effect on Object Detectors (YOLOv3)

Legend

- > 0.8
- 0.6 – 0.8
- < 0.6

Average Confidence – Morning

Height	Distance	Angle							
		0°	45°	90°	135°	180°	225°	270°	315°
0.6m	2.5m	0.98	0.66	0.52	0.78	0.95	0.54	0.55	0.91
	5m	0.91	0.78	0.70	0.88	0.71	0.83	0.81	0.41
	10m	0.43	0.46	0.00	0.34	0.46	0.35	0.33	0.50
1.8m	2.5m	0.97	0.98	0.92	0.99	0.99	0.97	0.92	0.97
	5m	0.85	0.85	0.94	0.97	0.95	0.90	0.88	0.97
	10m	0.63	0.39	0.00	0.31	0.41	0.44	0.00	0.33
2.4m	2.5m	1.0	0.99	0.96	0.99	1.0	0.99	0.99	1.0
	5m	0.90	0.90	0.68	0.66	0.94	0.99	0.96	0.98
	10m	0.96	0.62	0.44	0.86	0.80	0.60	0.57	0.76

Average Confidence – Afternoon

Height	Distance	Angle							
		0°	45°	90°	135°	180°	225°	270°	315°
0.6m	2.5m	0.96	0.59	0.65	0.72	0.88	0.94	0.84	0.86
	5m	0.79	0.78	0.77	0.50	0.63	0.31	0.35	0.31
	10m	0.45	0.37	0.28	0.48	0.71	0.28	0.27	0.30
1.8m	2.5m	0.94	0.96	0.89	0.95	0.96	0.97	0.81	0.91
	5m	0.96	0.66	0.71	0.66	0.78	0.91	0.94	0.98
	10m	0.63	0.59	0.39	0.68	0.67	0.29	0.42	0.36
2.4m	2.5m	1.0	0.99	0.82	0.99	0.99	0.99	0.95	0.99
	5m	0.89	0.91	0.67	0.99	0.96	0.96	0.86	0.88
	10m	0.62	0.52	0.43	0.49	0.63	0.51	0.34	0.51

Average Confidence – Night

Height	Distance	Angle							
		0°	45°	90°	135°	180°	225°	270°	315°
0.6m	2.5m	0.88	0.55	0.27	0.77	0.88	0.75	0.41	0.42
	5m	0.99	0.99	0.80	0.96	0.99	0.99	0.92	0.99
	10m	0.72	0.77	0.62	0.73	0.97	0.92	0.77	0.77
1.8m	2.5m	0.92	0.40	0.35	0.81	0.97	0.63	0.35	0.83
	5m	0.99	0.98	0.58	0.54	0.99	0.99	0.91	0.98
	10m	0.94	0.80	0.80	0.79	0.99	0.87	0.72	0.81
2.4m	2.5m	0.99	0.71	0.38	0.84	0.99	0.86	0.77	0.91
	5m	0.99	0.94	0.46	0.59	0.98	0.97	0.83	0.85
	10m	0.98	0.73	0.68	0.87	0.91	0.65	0.90	0.99

Analysis – Distance/Angle Effect on Object Detectors (Faster R-CNN)

Legend
- > 0.8
- 0.6 – 0.8
- < 0.6

Average Confidence - Lab

Height	Distance	Angle							
		0°	45°	90°	135°	180°	225°	270°	315°
0.6m	2.5m	1.0	0.95	0.91	0.92	0.93	0.91	0.81	0.94
	5m	0.68	0.69	0.82	0.62	0.96	0.65	0.79	0.67
	10m	0.57	0.58	0.99	1.0	0.99	0.63	0.72	0.59
1.8m	2.5m	1.0	1.0	0.99	1.0	1.0	0.92	0.99	1.0
	5m	0.76	0.72	0.85	0.66	1.0	0.69	0.77	0.75
	10m	0.77	0.63	0.61	0.56	0.82	0.92	0.64	0.80
2.4m	2.5m	1.0	0.99	0.98	0.99	0.98	0.99	0.98	1.0
	5m	1.0	0.68	0.78	0.78	1.0	0.80	0.80	1.0
	10m	0.82	0.60	0.59	0.83	0.84	0.79	0.55	0.97

Average Confidence - Morning

Height	Distance	Angle							
		0°	45°	90°	135°	180°	225°	270°	315°
0.6m	2.5m	0.99	1.0	0.99	1.0	1.0	1.0	1.0	1.0
	5m	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	10m	0.96	0.66	0.53	0.54	0.61	0.99	1.0	0.60
1.8m	2.5m	0.99	0.99	0.99	0.99	0.99	0.99	0.98	0.99
	5m	0.61	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	10m	0.88	0.97	0.84	0.89	0.90	0.90	0.91	0.86
2.4m	2.5m	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	5m	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	10m	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0

Average Confidence - Afternoon

Height	Distance	Angle							
		0°	45°	90°	135°	180°	225°	270°	315°
0.6m	2.5m	0.99	1.0	0.99	0.99	0.99	0.99	0.99	1.0
	5m	0.99	0.99	0.99	0.99	0.99	0.99	1.0	0.99
	10m	0.95	0.94	0.93	0.94	0.95	0.92	0.94	0.95
1.8m	2.5m	0.99	0.99	0.99	0.98	0.99	0.99	0.99	0.99
	5m	0.98	0.94	0.91	0.99	0.97	0.90	0.95	0.98
	10m	0.86	0.86	0.77	0.85	0.86	0.80	0.85	0.82
2.4m	2.5m	1.0	0.99	1.0	0.99	0.99	0.99	0.99	0.99
	5m	0.90	0.92	0.91	0.97	0.93	0.90	0.91	0.88
	10m	0.73	0.71	0.64	0.71	0.77	0.71	0.66	0.67

Average Confidence - Night

Height	Distance	Angle							
		0°	45°	90°	135°	180°	225°	270°	315°
0.6m	2.5m	0.97	0.86	0.72	0.90	0.94	0.96	0.70	0.93
	5m	1.0	1.0	0.70	0.80	1.0	1.0	0.64	1.0
	10m	1.0	1.0	0.80	0.90	1.0	1.0	0.76	0.84
1.8m	2.5m	0.94	0.88	0.82	0.86	0.88	0.79	0.65	0.90
	5m	0.78	0.78	0.53	0.76	1.0	0.68	0.57	0.99
	10m	0.85	0.68	0.52	0.72	0.84	0.74	0.98	0.96
2.4m	2.5m	1.0	0.70	0.62	0.75	1.0	0.69	0.65	0.76
	5m	0.95	0.78	0.54	0.94	0.87	0.96	0.99	0.95
	10m	0.94	0.74	0.70	0.96	0.95	0.80	0.70	0.88

Analysis – Distance/Angle Effect on Object Detectors

Conclusions/Takeaways:

- The confidence of an object detector is dependent on the:
  - **Distance** of the person from the object detector.
  - **Angle** of the person relative to the object detector.
  - **Height** of the object detector.
- The positioning of a person in a frame directly influences an object detector's confidence when detecting the person.
- Since object detectors have a minimum confidence threshold, a person's position may cause them to evade detection by an object detector.

Analysis – Distance/Angle Effect on Object Detectors

We repeated the previous experimental setup, which utilized a Full HD 1080p camera, twice, utilizing:

- An HD 720p camera.
- An SD 480p camera.

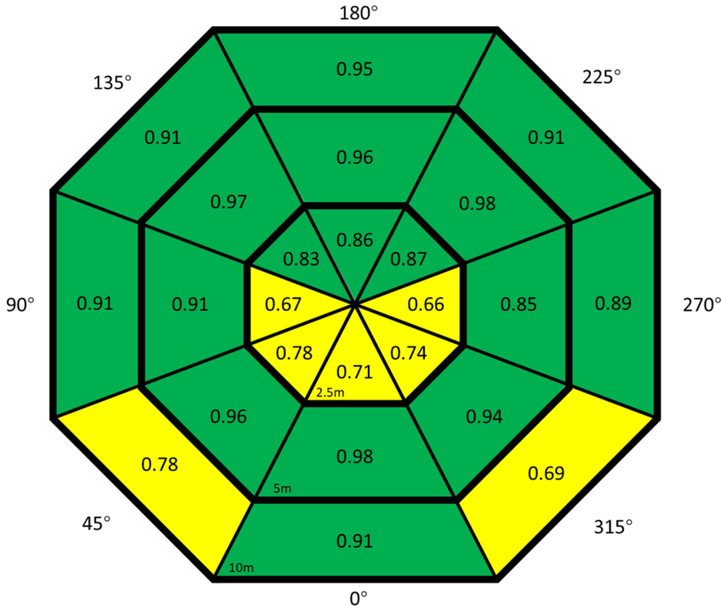
Our objective was to determine whether the previously obtained results are independent of camera resolution.

Analysis – Distance/Angle Effect on Object Detectors (YOLOv3)

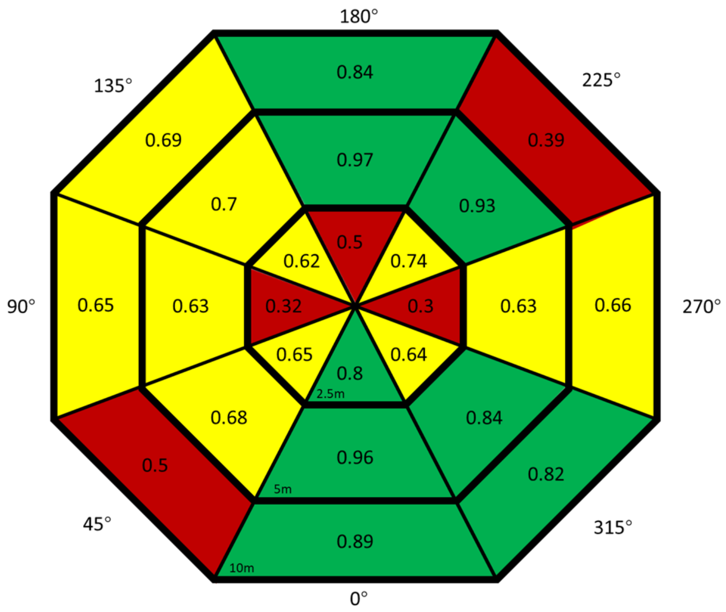
Results (Camera Resolution: 720x1280):

Legend
- > 0.8
- 0.6 – 0.8
- < 0.6

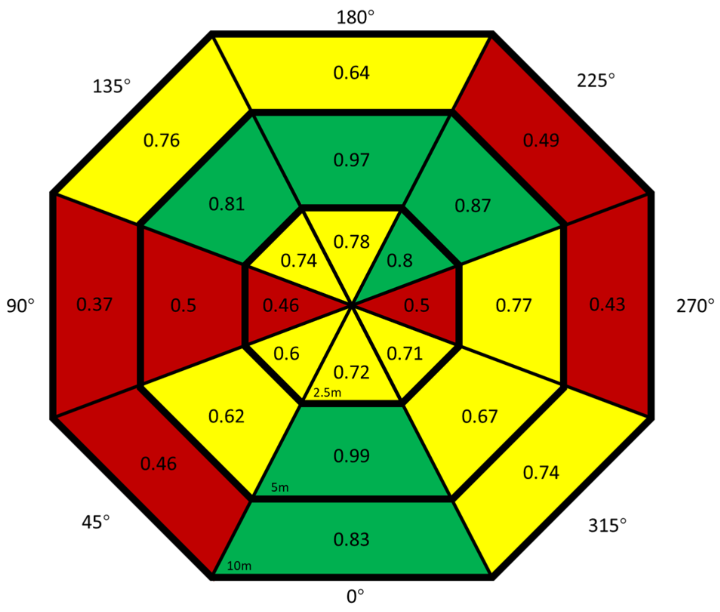
Height: 0.6m



1.8m



2.4m

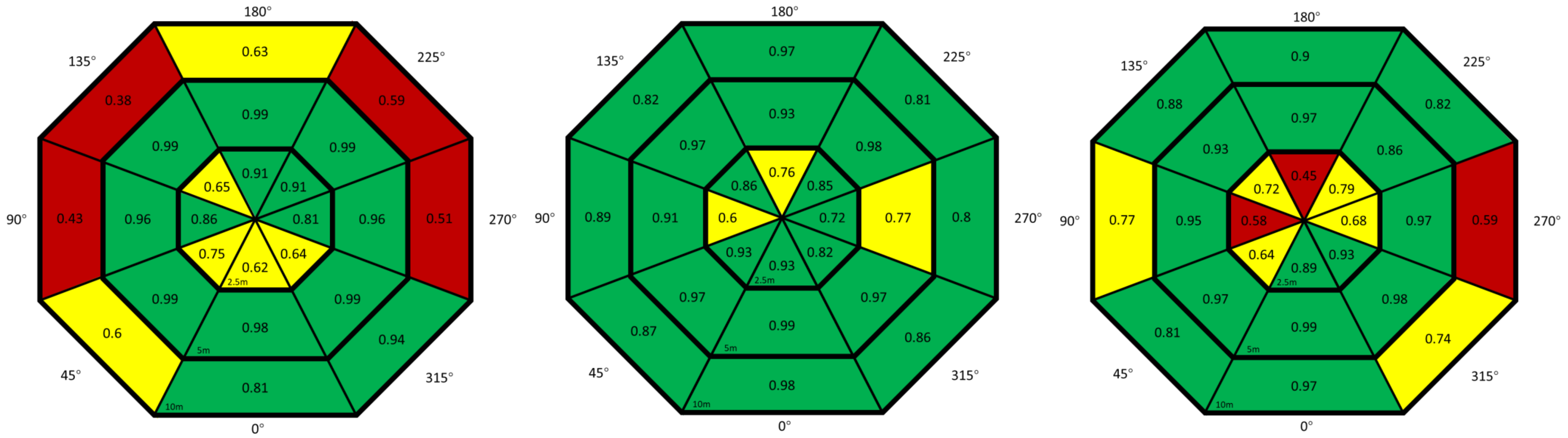


Analysis – Distance/Angle Effect on Object Detectors (YOLOv3)

Results (Camera Resolution 480x640):

Legend
- > 0.8
- 0.6 – 0.8
- < 0.6

Height: 0.6m 1.8m 2.4m



Analysis – Distance/Angle Effect on Object Detectors

Conclusions/Takeaways:

The camera resolution:

- Effects which distances/angles/heights an object detector demonstrate high or low confidence.
- Does not effect the observed phenomena of varying object detector confidence given different distances/angles/heights.

Analysis – Location Effect on Object Detectors

Experimental Setup:

We utilized public recordings of pedestrian traffic in Shibuya Crossing.

- We took footage from six consecutive 4 hour recordings (24 hours).
- We fed one frame from each two seconds of footage to an object detector.

For each pixel in the frame, we calculated the mean confidence of the object detector's "person" detection bounding boxes that cover the pixel.

Photo by Sho Studio on Unsplash

Analysis – Location Effect on Object Detectors

We repeated this analysis with five different object detectors:

YOLOv3, Faster R-CNN, SSD, DiffusionDet, RTMDet.

We repeated this analysis in three different locations:

- Shibuya Crossing, Tokyo.
- Broadway, New York.
- Castro Street, San Francisco.

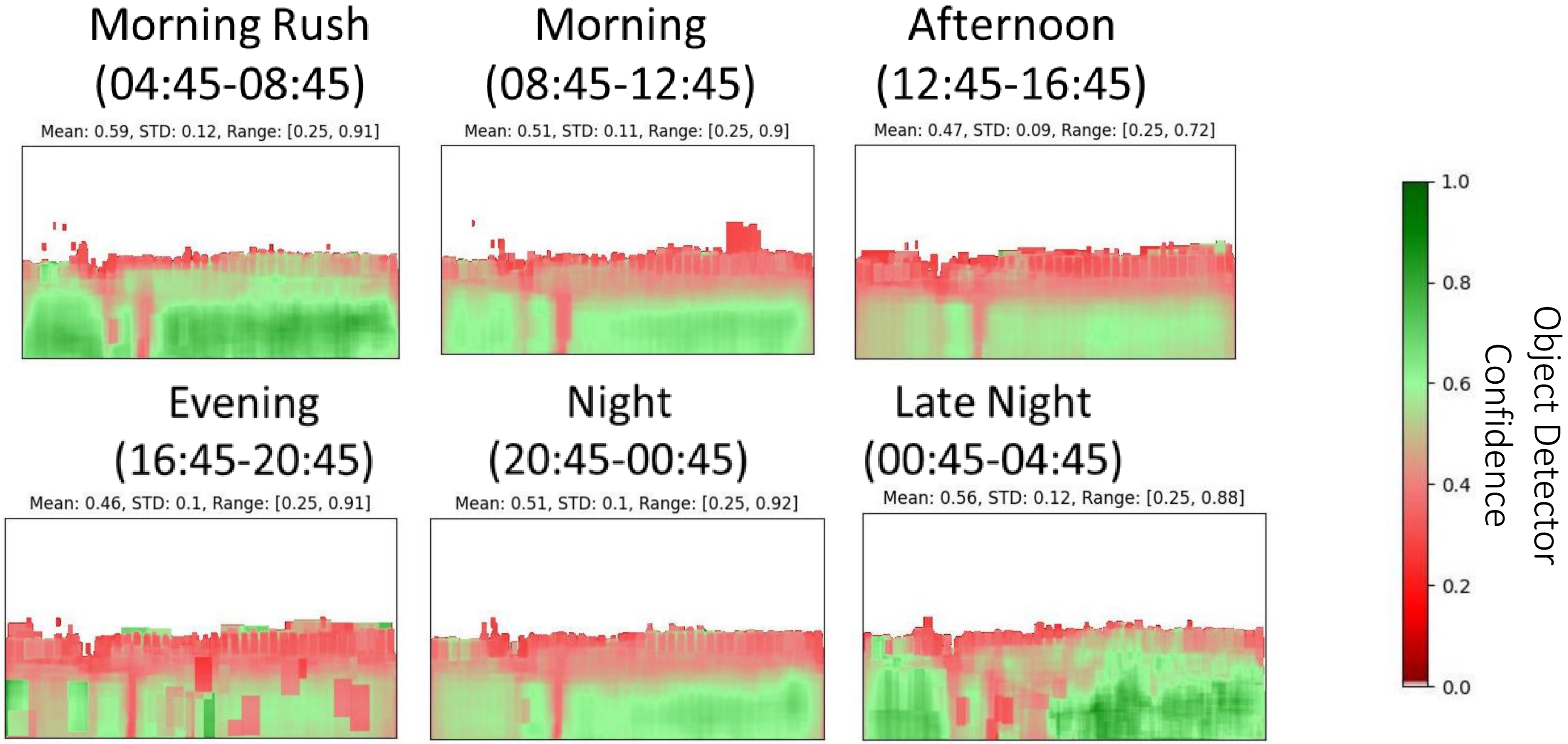
Photo by Sho Studio on Unsplash

Photo by RAMSHA ASAD on Unsplash

Photo by Piotr Musioł on Unsplash

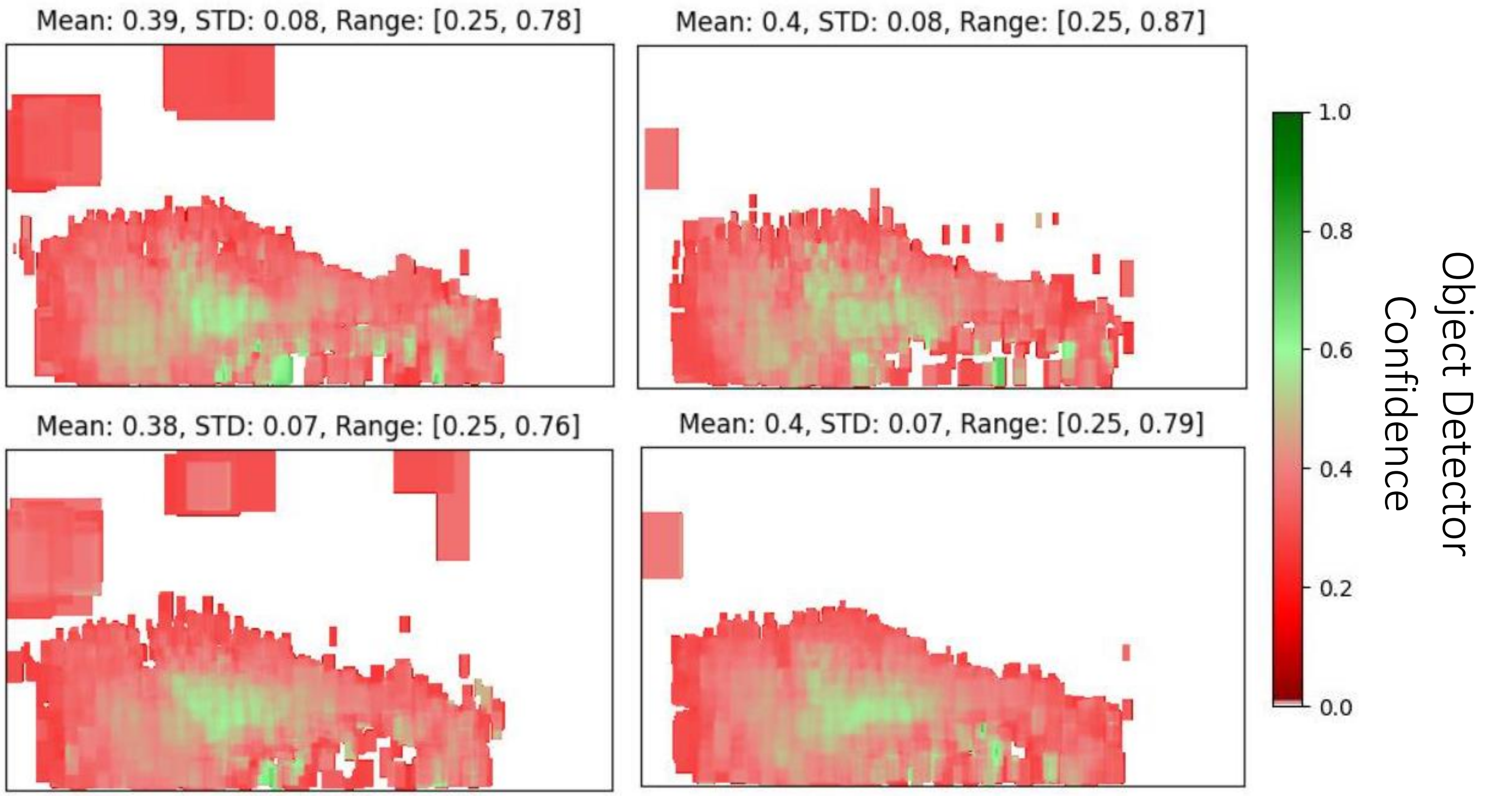
Analysis – Location Effect on Object Detectors (YOLOv3)

Results (Shibuya):



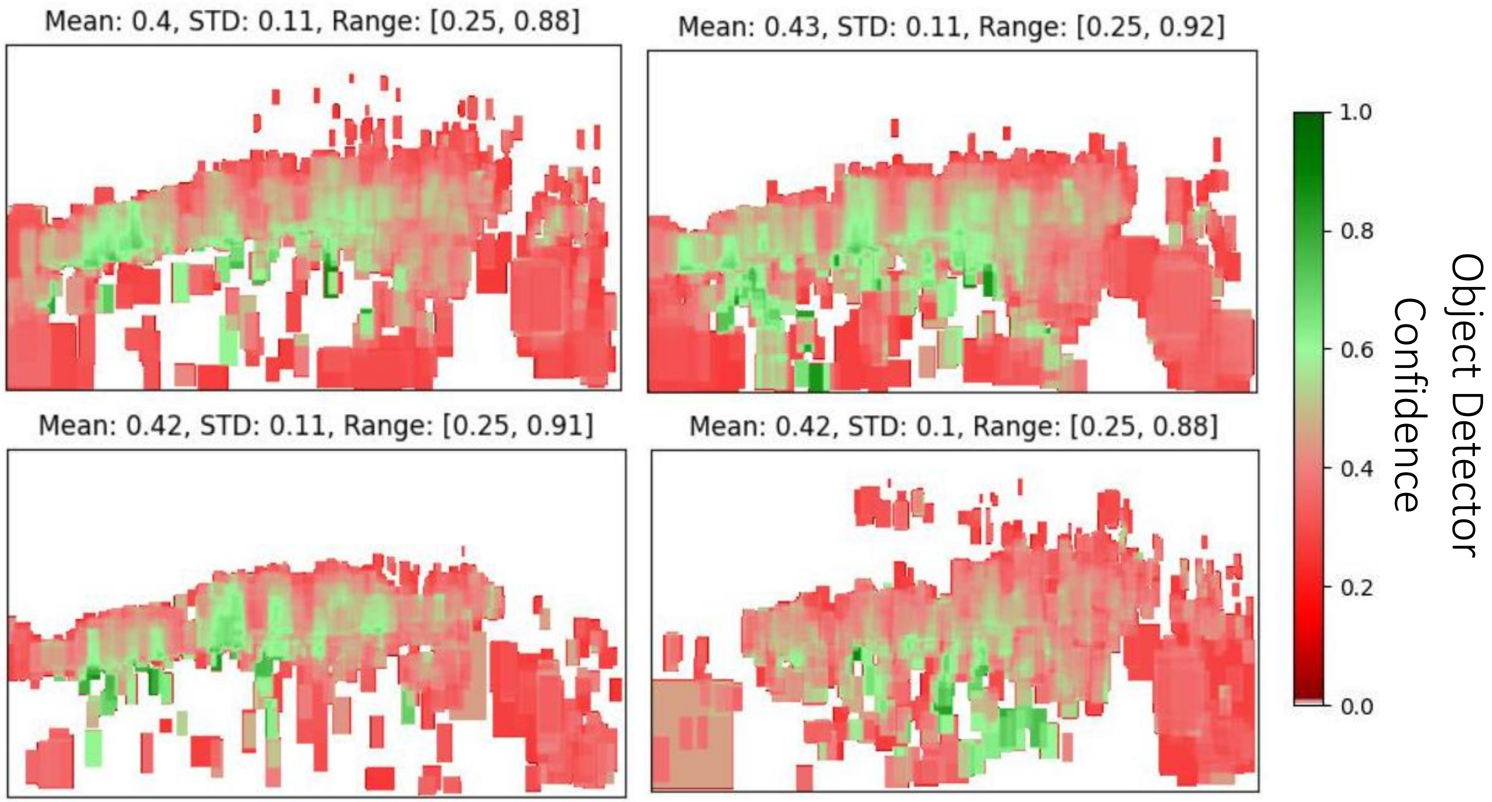
Analysis – Location Effect on Object Detectors (YOLOv3)

Results (Broadway):



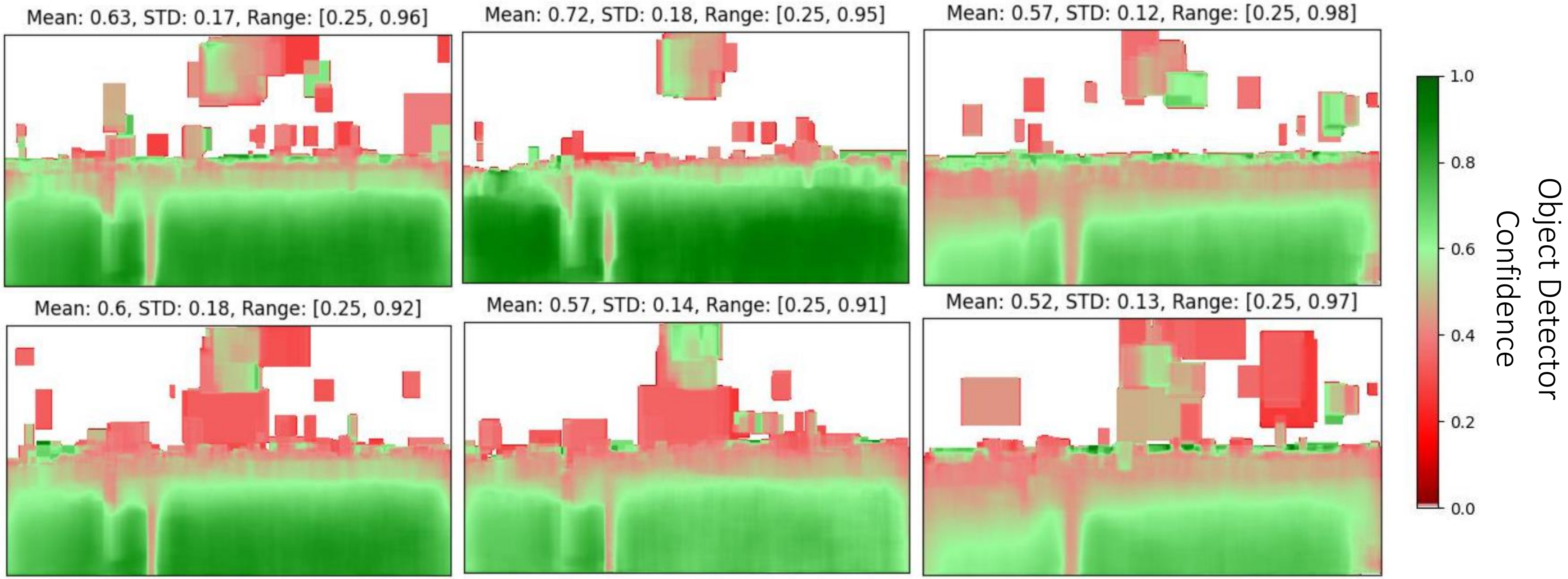
Analysis – Location Effect on Object Detectors (YOLOv3)

Results (Castro Street):



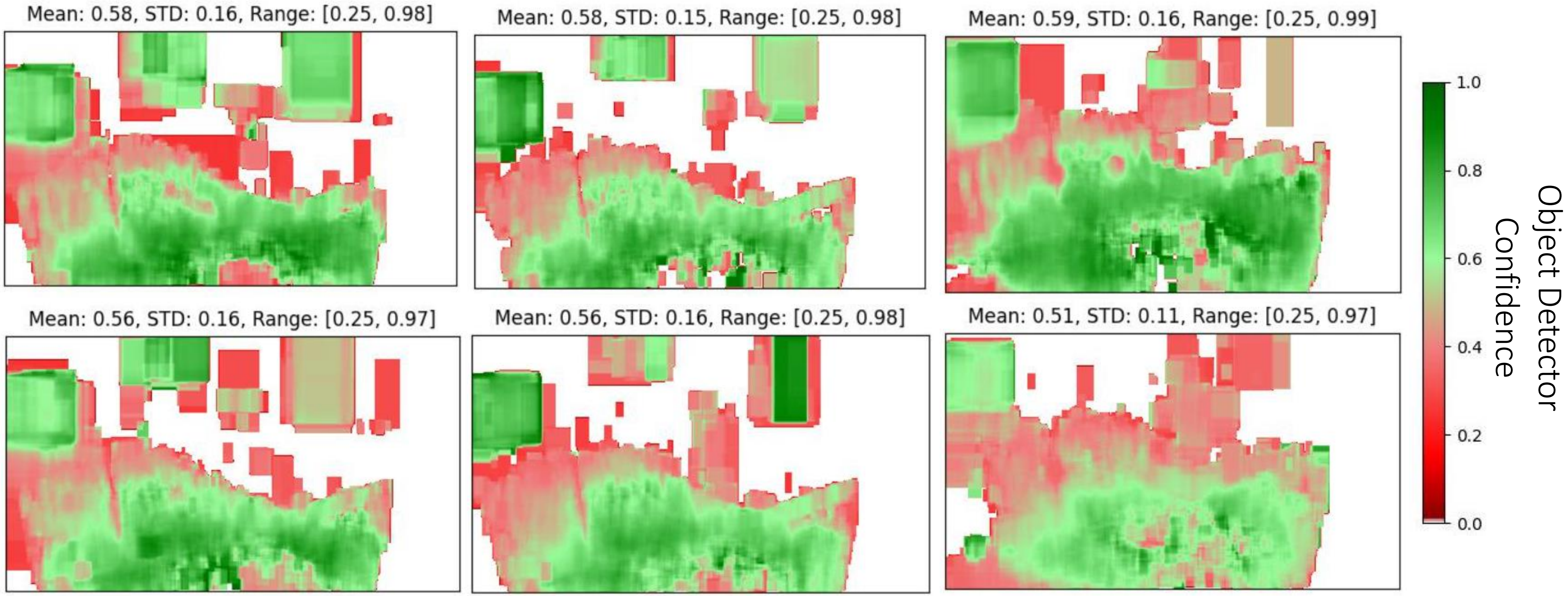
Analysis – Location Effect on Object Detectors (Faster R-CNN)

Results (Shibuya):



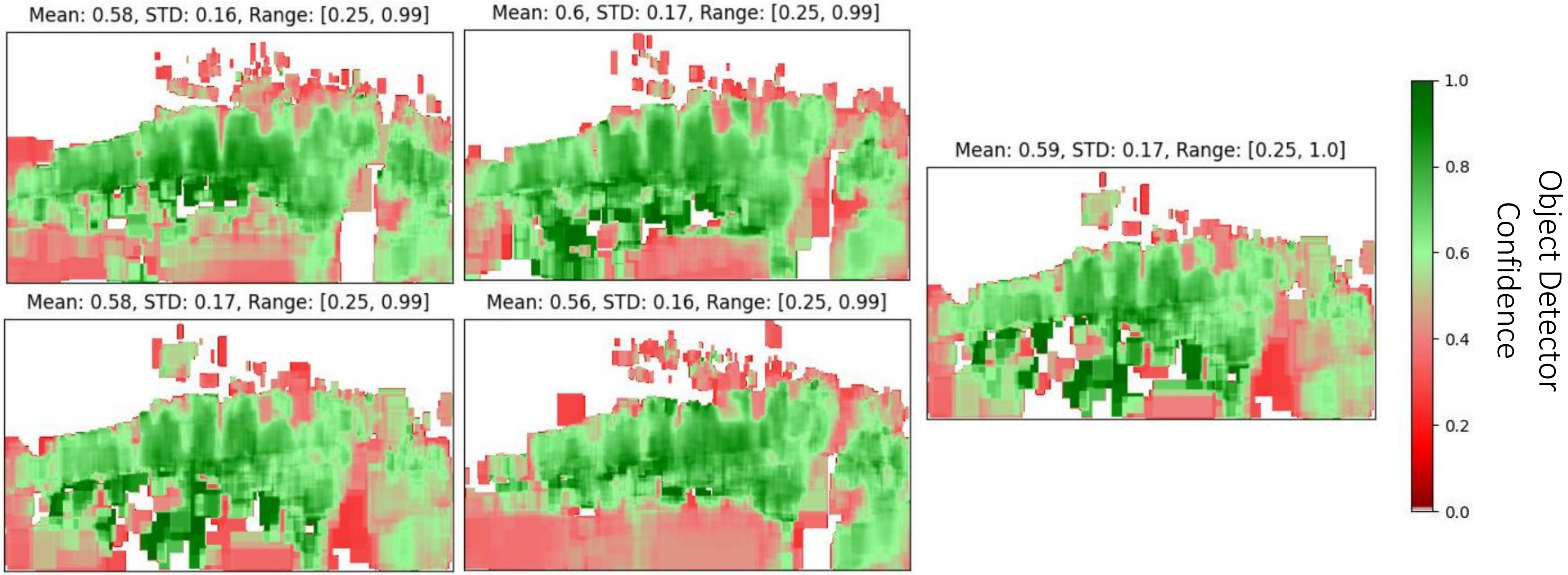
Analysis – Location Effect on Object Detectors (Faster R-CNN)

Results (Broadway):



Analysis – Location Effect on Object Detectors (Faster R-CNN)

Results (Castro Street):



Analysis – Location Effect on Object Detectors

Conclusions/Takeaways:

- The average confidence of an object detector at detecting people is dependent on a person's location in the frame.
- In a given scene/viewing point, there are areas the object detector is stronger and weaker at detecting people.
- These "strong" and "weak" areas can be viewed in a heatmap, giving an observer an overview of the object detector's person detection confidence in a given area.
- These findings are applicable to various locations and object detection models.

TipToe – Objective

Given knowledge of/access to:

- Varying confidence levels with respect to distance, angle, and height.
- An object detector’s heatmap of strong and weak detection areas.

Find a path across the scene where an attacker/pedestrian is either:

- Detected with minimal confidence, or
- Not detected.

Terminology and Problem Definition

Terminology:

- Scene – a physical world area observed by an object detector via camera stream/recording.
- Confidence Heatmap – for each pixel in the scene, the average confidence of the bounding boxes which fall on the pixel.
- Detection Heatmap – for each pixel in the scene, the number of bounding boxes which fall on the pixel.

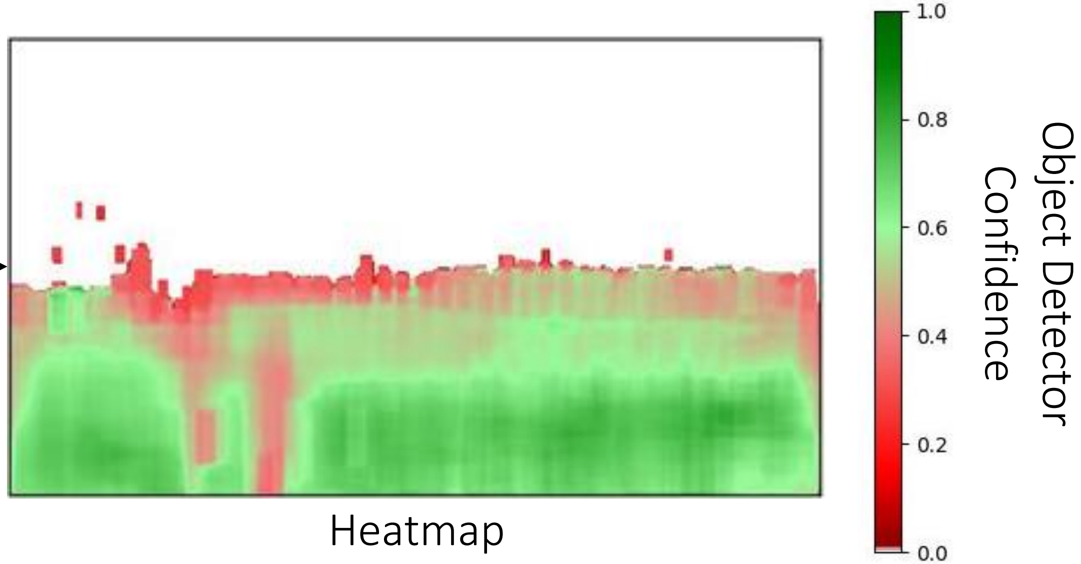
Terminology and Problem Definition

Problem Definition:

The location-based evasion attack leverages confidence measurements recorded in the confidence heatmap.

Each cell [i,j] in the confidence heatmap records the average confidence of ‘person’ bounding boxes falling on pixel [i,j] in the observed scene.

Scene



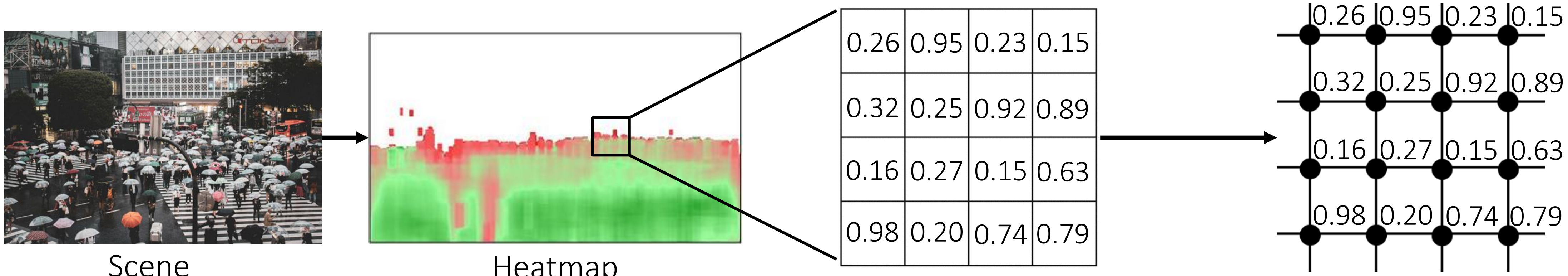
Heatmap

Terminology and Problem Definition

Problem Definition:

We performed a reduction of the scene/heatmap from a pixel-based matrix representation to a graph-based representation.

Each pixel is represented by a node in the graph-based representation, with two-way edges connecting each node to its cardinal (non-diagonal) neighbors).



Scene

Heatmap

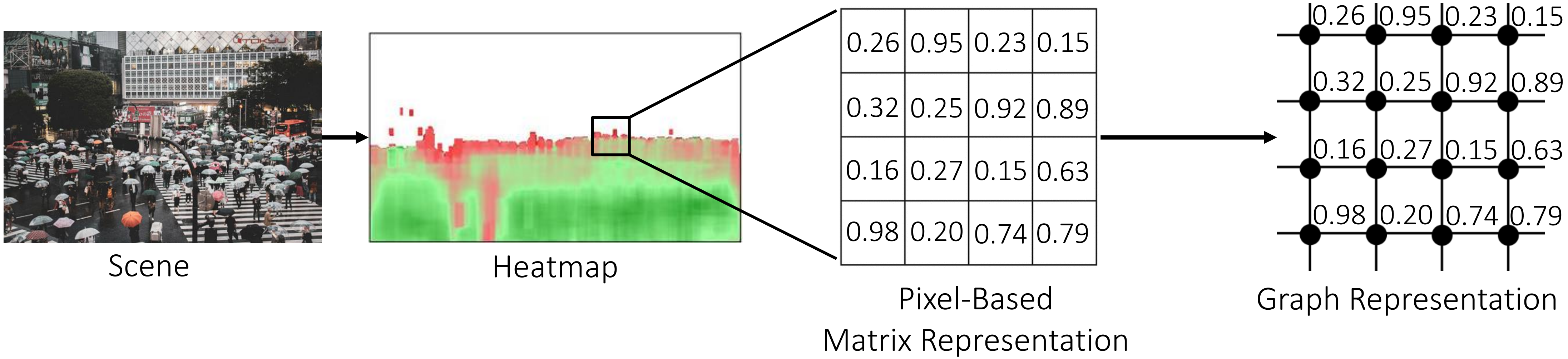
Pixel-Based Matrix Representation

Graph Representation

Terminology and Problem Definition

Problem Definition:

Each node in the graph representation is defined a weight equal to the confidence level recorded in the confidence heatmap's respective pixel.

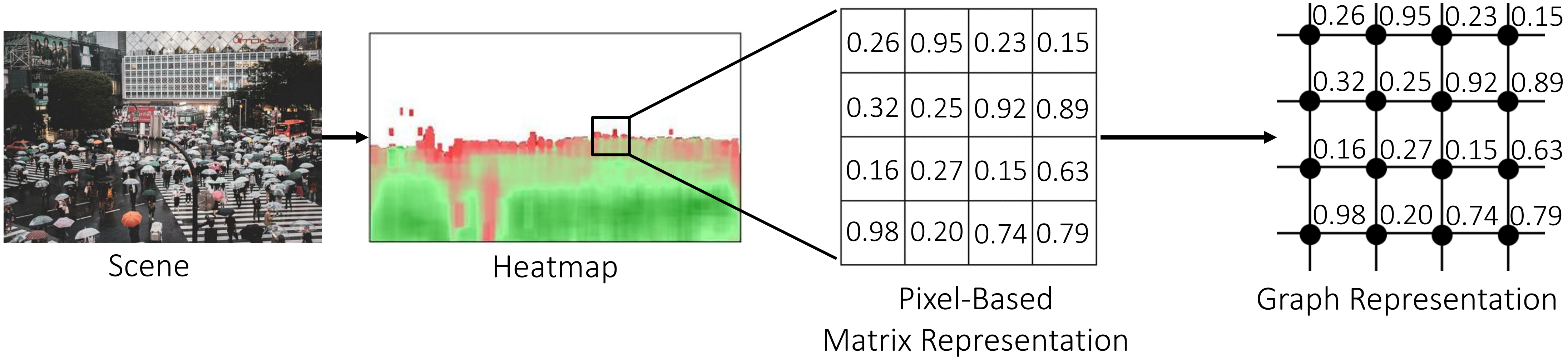


Terminology and Problem Definition

Problem Definition:

We reduce the problem of finding an optimal path between two points in the scene to finding an optimal path between two nodes in the graph representation.

In order to avoid detection, an attacker would optimize: 1) the maximum confidence on the path, and 2) the average confidence of the nodes on the path.



Threat Model

Attacker’s goal: construct a path between two points in a scene observed by an AI-based pedestrian detector with minimal/no detection.

The attacker has access to:

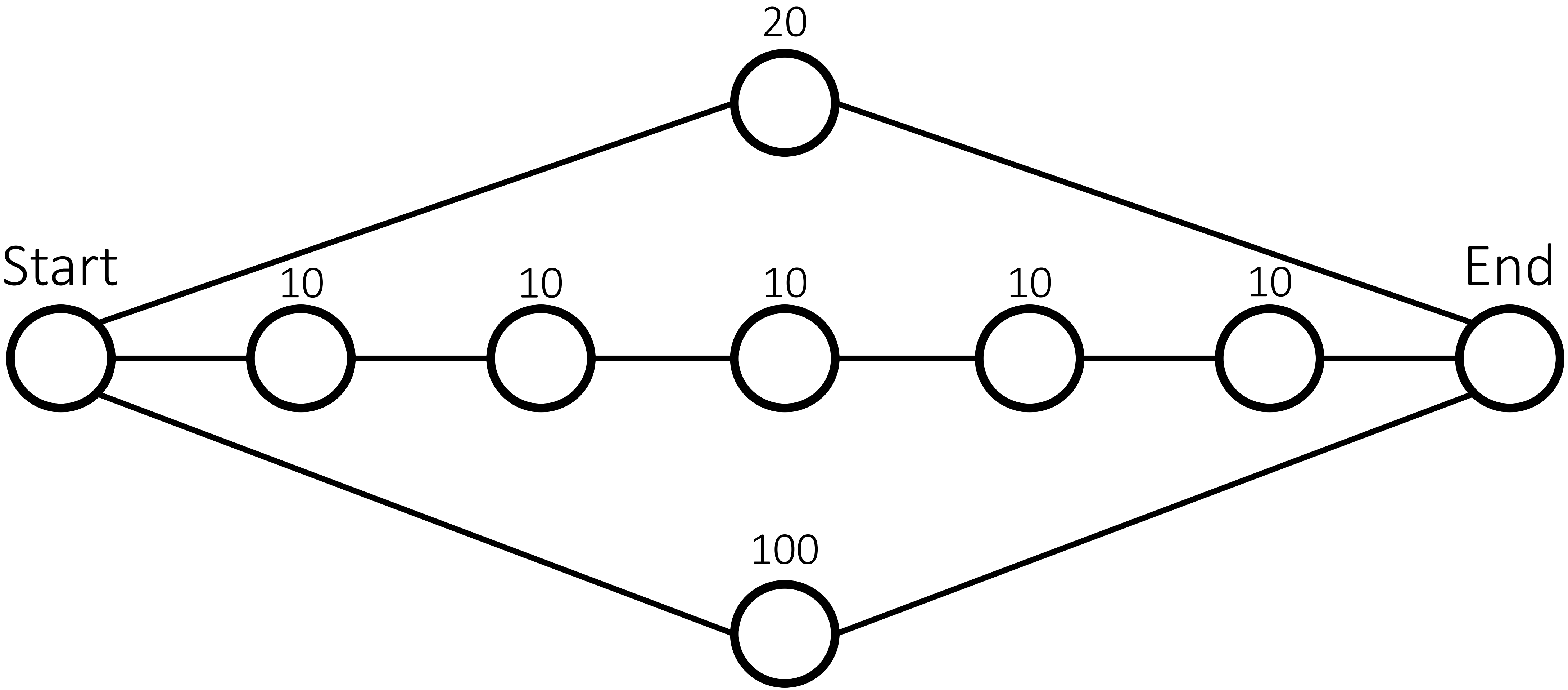
- A confidence heatmap of the pedestrian detector.
- A detection heatmap of the pedestrian detector.

The attacker doesn’t have access to:

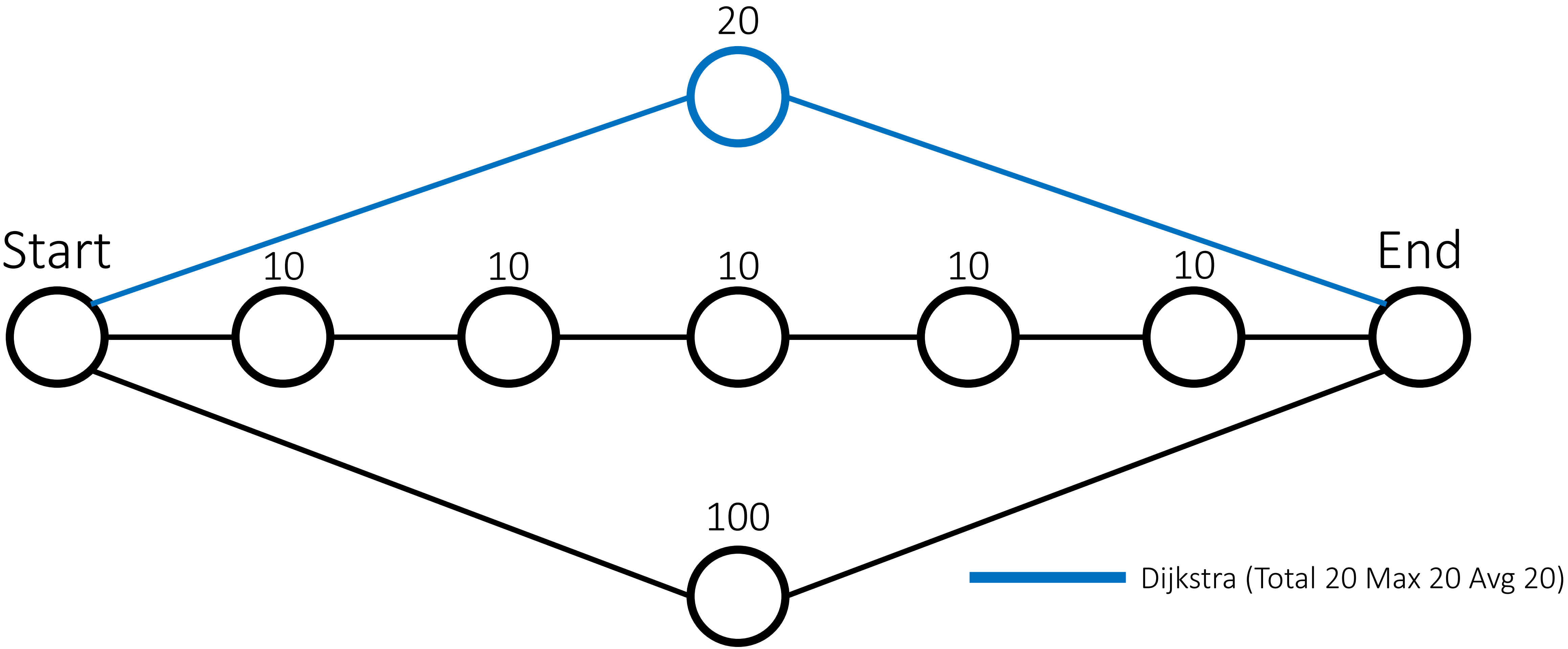
- Sophisticated adversarial perturbation methods.
- Machinery to directly interfere with the pedestrian detector’s perception of the scene.

•The only factor controlled by the attacker is the evading person’s location in the scene and the path they traverse through the scene.

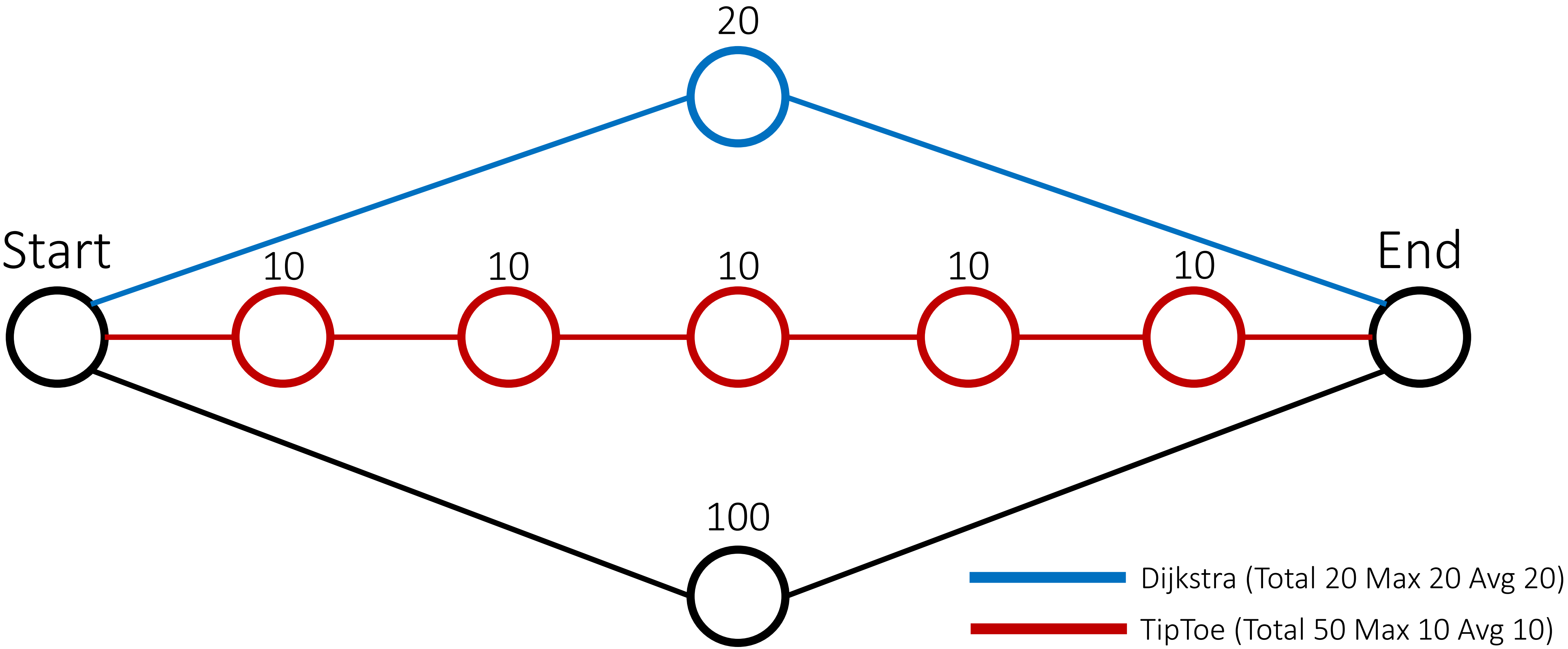
TipToe Location-Based Evasion Attack



TipToe Location-Based Evasion Attack



TipToe Location-Based Evasion Attack



TipToe Location-Based Evasion Attack

Input: H_c, H_d, s, e.
Output: p - lowest-max-cost path between s and e.

$H_a \leftarrow H_c[(i,j) \in H_c | H_d(i,j) > 0]$.
$Q \leftarrow$ priority queue.
for $p \in H_a$ **do**
  $dist(p) \leftarrow \infty$.
  $prev(p) \leftarrow$ UNDEFINED.
  $Q.add(p)$, priority defined by $dist(p)$.
$dist(s) \leftarrow 0$
while Q is not empty: **do**
  $u \leftarrow$ node in Q with min $dist(u)$.
  $Q.remove(u)$
  for $\{v \in Q | v$ is a neighbor of $u\}$: **do**
    $alt \leftarrow min(max(dist(u), H_c(v)), dist(v))$.
    if $alt < dist(v)$ **then**
      $dist(v) \leftarrow alt$.
      $prev(v) \leftarrow u$.
return: $dist, prev$

Evaluation - TipToe

Experimental Setup:

- Data – Six four-hour long videos of footage from Shibuya Crossing (24 consecutive hours in total).
- We generated confidence and detection heatmaps for each video.
- We took 10 randomly-chosen start and end points on the scene.
- On the set of 100 paths between each pair of start and end points, we calculated the average:
  - Max node confidence of the path.
  - Average per-step confidence of the path.
- We calculated these values for three path-finding strategies: 1) TipToe, 2) a random Manhattan-distance path, and 3) a random length path.

Evaluation – TipToe (YOLOv3)

Results (Shibuya):

Video	Max Confidence (Attack)	Max Confidence (Man. Dist.)	Max Confidence (Random)	Average Confidence (Attack)	Average Confidence (Man. Dist.)	Average Confidence (Random)
Morning Rush (4:45-8:45)	0.65	0.69	0.70	0.44	0.56	0.60
Morning (8:45-12:45)	0.59	0.60	0.60	0.39	0.50	0.53
Afternoon (12:45-16:45)	0.52	0.55	0.55	0.37	0.44	0.48
Evening (16:45-20:45)	0.53	0.60	0.63	0.40	0.45	0.47
Night (20:45-0:45)	0.58	0.65	0.62	0.40	0.46	0.49
Witching (0:45-4:45)	0.67	0.72	0.75	0.45	0.56	0.59

Evaluation – TipToe (YOLOv3)

Results (Broadway):

Video	Max Confidence (Attack)	Max Confidence (Man. Dist.)	Max Confidence (Random)	Average Confidence (Attack)	Average Confidence (Man. Dist.)	Average Confidence (Random)
Day 1	0.47	0.56	0.56	0.35	0.42	0.44
Day 2	0.45	0.53	0.53	0.33	0.40	0.41
Day 3	0.48	0.58	0.58	0.35	0.42	0.43
Night 1	0.45	0.56	0.54	0.34	0.38	0.40
Night 2	0.41	0.53	0.49	0.34	0.40	0.41
Night 3	0.40	0.50	0.51	0.34	0.38	0.39

Results (Castro):

Video	Max Confidence (Attack)	Max Confidence (Man. Dist.)	Max Confidence (Random)	Average Confidence (Attack)	Average Confidence (Man. Dist.)	Average Confidence (Random)
Sunrise	0.57	0.73	0.69	0.36	0.46	0.47
Day	0.49	0.60	0.64	0.36	0.42	0.45
Sunset	0.56	0.70	0.71	0.37	0.45	0.45
Night 1	0.57	0.66	0.65	0.39	0.48	0.49
Night 2	0.59	0.68	0.68	0.39	0.47	0.49

Evaluation – TipToe (Faster R-CNN)

Results (Shibuya):

Video	Max Confidence (Attack)	Max Confidence (Man. Dist.)	Max Confidence (Random)	Average Confidence (Attack)	Average Confidence (Man. Dist.)	Average Confidence (Random)
Day 1	0.74	0.83	0.79	0.49	0.62	0.64
Day 2	0.83	0.85	0.85	0.52	0.70	0.75
Day 3	0.66	0.71	0.69	0.51	0.58	0.60
Night 1	0.73	0.74	0.75	0.44	0.59	0.64
Night 2	0.63	0.67	0.68	0.44	0.52	0.57
Night 3	0.67	0.71	0.73	0.43	0.57	0.61

Results (Broadway):

Video	Max Confidence (Attack)	Max Confidence (Man. Dist.)	Max Confidence (Random)	Average Confidence (Attack)	Average Confidence (Man. Dist.)	Average Confidence (Random)
Day 1	0.73	0.82	0.83	0.47	0.57	0.62
Day 2	0.74	0.77	0.78	0.48	0.59	0.63
Day 3	0.72	0.80	0.80	0.50	0.56	0.61
Night 1	0.74	0.80	0.80	0.45	0.58	0.59
Night 2	0.68	0.74	0.75	0.43	0.54	0.58
Night 3	0.60	0.69	0.68	0.41	0.47	0.49

Results (Castro):

Video	Max Confidence (Attack)	Max Confidence (Man. Dist.)	Max Confidence (Random)	Average Confidence (Attack)	Average Confidence (Man. Dist.)	Average Confidence (Random)
Sunrise	0.77	0.88	0.88	0.50	0.67	0.68
Day	0.69	0.87	0.87	0.49	0.64	0.64
Sunset	0.71	0.85	0.86	0.50	0.63	0.64
Night 1	0.77	0.91	0.91	0.53	0.65	0.66
Night 2	0.69	0.82	0.82	0.50	0.60	0.60

Evaluation

Conclusions/Takeaways:

Compared to direct (Manhattan distance) and random paths, paths generated by TipToe have lower:

- Maximum path confidence (up to ~16% reduction).
- Average path confidence (up to ~27% reduction).

These findings demonstrate that an attacker can manipulate the confidence level of an object detector by leveraging a person’s location as they traverse the scene.

Conclusions & Takeaways

Object detectors are sensitive to a person's position (distance, angle, height) in the frame:

- Reflected in the confidence score of the model's ”person” detections.
- Independent of model architecture, video quality, and time of day.

This behavior is also present in detections of pedestrian traffic in public areas:

- Reflected in varying confidence in different areas of the frame.
- Can be visualized in a "heatmap" of stronger and weaker detection areas.

The TipToe evasion attack can leverage these varying detection areas to construct minimum-confidence paths through a scene.

- Lower max and average confidence than direct and random paths through the scene.

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Thank you!

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